

HR Analytics Dashboard using Power BI

Project Overview

The HR Analytics Dashboard is a Business Intelligence project developed in Power BI to analyze employee data and identify key factors responsible for employee attrition within an organization.

This project transforms raw HR data into meaningful insights using interactive dashboards, visualizations, KPIs, and data analysis techniques. The dashboard helps HR managers and business leaders make data-driven decisions to improve employee retention, workforce planning, and organizational performance.

Business Problem

Employee attrition is one of the major challenges faced by organizations. High employee turnover leads to:

- Increased hiring and training costs
- Reduced productivity
- Loss of experienced employees
- Poor workforce stability
- Lower organizational efficiency

The company needs a solution to monitor employee attrition trends and understand the reasons behind employee turnover based on various factors such as age, salary, education, gender, job role, and years at the company.

This project aims to solve these business problems through HR data analysis and visualization.

Project Objectives

The main objectives of this project are:

- Analyze employee attrition patterns
- Identify departments with high attrition
- Understand employee demographics
- Monitor salary distribution and its impact on attrition
- Track employee experience and retention
- Support strategic HR decision-making

Tools & Technologies Used

Tool	Purpose
Power BI	Dashboard Creation & Visualization
Excel / CSV	
Power Query	Data Cleaning & Transformation
DAX	Data Analysis & Calculations
Data Modeling	Relationship Management

Power BI Concepts Used

1. Data Cleaning (Power Query)

Performed data preprocessing operations such as:

- Removing null values
- Renaming columns
- Changing data types
- Removing duplicates
- Filtering unwanted data
- Creating conditional column

DAX Queries / Measures Used

Employee Count

```
```DAX
Employee Count = COUNT(Employee[EmployeeID])
```
```

Attrition Count

```
```DAX
```

Attrition Count = CALCULATE(COUNT(Employee[Attrition]), Employee[Attrition] = "Yes")

...

## Attrition Rate

```DAX

Attrition Rate =

DIVIDE([Attrition Count], [Employee Count], 0) * 100

...

Average Age

```DAX

Average Age = AVERAGE(Employee[Age])

...

## Average Salary

```DAX

Average Salary = AVERAGE(Employee[MonthlyIncome])

...

Average Years at Company

```DAX

Avg Years = AVERAGE(Employee[YearsAtCompany])

``

# Dashboard Analysis Performed

## 1. Employee Overview Analysis

- Total Employees

- Attrition Count

- Attrition Rate

- Average Age

- Average Salary
- Average Years at Company

### Business Insight:

Provides a complete overview of workforce statistics.

## ## 2. Attrition by Education

Analyzed employee attrition based on educational background such as:

- Life Sciences
- Medical
- Marketing
- Technical Degree

### Business Insight:

Identifies which education backgrounds have higher turnover rates.

## ## 3. Attrition by Age Group

Age groups analyzed:

- 18–25
- 26–35
- 36–45
- 46–55
- 55+

### Business Insight:

Helps identify which age group leaves the organization most frequently

## ## 4. Attrition by Gender

Comparison between:

- Male Employees

- Female Employees

### Business Insight:

Helps HR understand gender-based retention trends.

## ## 5. Attrition by Salary Slab

Salary categories analyzed:

- Up to 5K

- 5K–10K

- 10K–15K

- 15K+

### Business Insight:

Shows relationship between salary and employee turnover.

## ## 6. Attrition by Job Role

Analyzed attrition across job roles such as:

- Laboratory Technician

- Sales Executive

- Research Scientist

- Human Resources

### Business Insight:

Identifies high-risk job roles with maximum attrition.

## ## 7. Attrition by Years at Company

Analyzed employee attrition based on years spent in the organization.

### Business Insight:

Helps determine at which stage employees are more likely to leave.

## # Types of Analysis Performed

- Descriptive Analysis
- Employee Attrition Analysis
- Workforce Demographic Analysis
- Salary Analysis
- Department-wise Analysis
- Trend Analysis
- KPI Reporting
- HR Performance Analysis

## Key Features of the Dashboard

- Interactive Dashboard
- Dynamic Filtering
- Department-wise Analysis
- HR KPI Tracking
- Business Insights Visualization
- User-Friendly Interface
- Real-time Data Exploration

## # Business Impact

This dashboard helps organizations to:

- Reduce employee turnover
- Improve employee satisfaction
- Enhance workforce planning
- Identify retention issues
- Support HR strategic decisions
- Increase organizational productivity

## # Future Enhancements

- Predictive Attrition Analysis using Machine Learning
- Real-Time Database Connectivity

- Employee Performance Tracking
- AI-Based HR Recommendations
- Advanced HR Forecasting

# Dashboard Preview



# Conclusion

The HR Analytics Dashboard successfully converts raw HR data into actionable business insights using Power BI. This project demonstrates strong skills in data visualization, business intelligence, DAX calculations, and HR analytics.

The dashboard enables organizations to monitor employee attrition trends, improve retention strategies, and make smarter HR decisions using data-driven insights.

# Author

Gaurav yadav

Power BI Developer | Data Analyst